ModelIt!

Student Activity Pack

# How Does Your Brain Know That's Hot?

The Lightning-Fast Science of Senses and Signals

Grade 4 | NGSS 4-LS1-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

**1 How do you think your body knows when something is hot or cold?**

**2 Why do you pull your hand away from something hot before you even think about it?**

**3 Draw a picture showing how a message might travel from your finger to your brain.**

**4 What are your five senses and what does each one detect?**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Heat Source Intensity | Nerve Signal Speed | Brain Response Time

| EXTERNAL | INTERNAL |
|---|---|
| | |
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Warm Cup

Heat Source Intensity set to low

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Hot Stove

Heat Source Intensity set to high

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Boiling Water

Heat Source Intensity set to maximum

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Pain Level** — How much pain the sensation causes — more pain triggers faster and stronger responses

**Nerve Pathway Length** — How far the signal has to travel — signals from your toes take longer to reach your brain than signals from your face

**Attention Level** — Whether you are paying attention — your brain can process signals faster when you are focused and alert

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**When you touch a hot stove, your hand pulls back before you even think about it — how does your body react SO fast?** Use evidence from your simulation to support your answers.

**1 Why does your body pull away from hot things even before you feel the pain?**

**2 What would happen if your nerves could not send signals to your brain?**

**3 Why do you think your reflexes are faster than your thinking?**

**4 How are nerves in your body similar to electrical wires in a building?**

**5 How did your model help you understand something happening inside your body that you cannot see?**

# 8 STEM Challenge: Design a Human Alarm System

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## YOUR ENGINEERING MISSION

Design a model of a body alarm system that detects a danger signal, sends it quickly, and triggers a response — just like your nervous system.

### THE SCENARIO

A children's hospital needs a teaching model to help young patients understand how their nervous system works. Your team must design a hands-on model that shows how a signal travels from a sensor to the brain and triggers a response. The faster the signal, the better!

### Engineering Design Thinking

1. WHAT MATERIAL WILL REPRESENT NERVES CARRYING THE SIGNAL (STRING, DOMINOES, TUBES)?
2. HOW WILL YOU SHOW THE SPEED DIFFERENCE BETWEEN A WEAK SIGNAL AND A STRONG DANGER SIGNAL?
3. WHAT WILL REPRESENT THE BRAIN MAKING A DECISION?
4. HOW WILL YOU SHOW THE RESPONSE (MUSCLES MOVING) AT THE END OF THE CHAIN?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Neuroscientists and Neurologists study how the brain and nervous system work, helping people with brain injuries, nerve disorders, and sensory problems. They earn $80,000–$150,000/year.